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Feature Engineering and Model Optimization Techniques
Practice Assignment • 15 min

Due Nov 9, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item

Instructions

1. In a recent study, the implementation of AI in network sensors resulted in a reduction of false positives by what percentage range? **1 / 1 point**

- [30, 40)
- [70, 80)
- [50, 60) ✓
- [40, 50)
- [60, 70)

Nice work

Correct! The reduction in false positives typically falls within this range due to AI implementation.

2. Which of the following are crucial data quality considerations for AI models in cybersecurity? (Select all that apply) **1 / 1 point**

- [ ] Timeliness
- [ ] Volume
- [x] Accuracy

Nice work

Correct! Accuracy is essential for ensuring that the data used for AI models is correct and reliable.

- [ ] Redundancy
- [ ] Aesthetics
- [x] Consistency

Nice work

Correct! Consistency ensures that data remains reliable and usable for AI models.

- [x] Completeness

Nice work

Correct! Completeness ensures that the data encompasses all necessary aspects for effective AI modeling.

3. What is the term used for the process of prioritizing alerts in cybersecurity using AI? **1 / 1 point**

- [x] Triage

Nice work

Correct! The process used for prioritizing alerts is known as triage.

- [x] Triaging

Nice work

Correct! The process can also be referred to as triaging.

- [ ] Filtering
- [ ] Prioritization

4. Imagine you are developing an AI system for network security. During testing, you notice that the AI frequently misclassifies benign traffic as malicious. What is a likely cause of this issue? **1 / 1 point**

- The training data may be imbalanced, with more examples of malicious traffic than benign traffic. ✓
- The AI system is processing data too slowly, causing delays in classification.
- The AI model is too simple and lacks the complexity to differentiate between benign and malicious traffic.
- The network sensors are not functioning correctly, affecting data collection.

Nice work

Correct! An imbalanced training dataset can cause the AI to misclassify benign traffic as malicious.

5. Which of the following challenges is specifically associated with using AI in network sensors for cybersecurity? **1 / 1 point**

- Real-time processing requirements
- Difficulty in software updates
- High cost of hardware
- Handling encrypted traffic ✓
- Integration with legacy systems
- Lack of data storage options

Nice work

Correct! One major challenge for AI in network sensors is dealing with encrypted traffic, which limits the ability to analyze data effectively.